

Лабораторная работа №9

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

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Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков установки POP3/IMAP-сервера

1.1 Цель работы

- Приобретение практических навыков установки POP3/IMAP-сервера
- Освоение базовой конфигурации почтового сервиса

Раздел 2

2. Выполнение лабораторной работы

2.1 Установка Dovecot

- Переход в режим суперпользователя (`sudo -i`)

```
[root@server.bahis.net ~]# dnf -y install dovecot telnet
Last metadata expiration check: 4:06:08 ago on Wed 11 Feb 2026 07:26:23 PM UTC.
Dependencies resolved.
=====
Package                Arch      Version                               Repository      Size
=====
Installing:
dovecot                 x86_64    1:2.3.16-15.el9                     appstream       4.7 M
telnet                  x86_64    1:0.17-85.el9                       appstream        63 k
Installing dependencies:
clucene-core            x86_64    2.3.3.4-42.20130812.e8e3d20git.el9  appstream       585 k
libexttextcat           x86_64    3.4.5-11.el9                        appstream       209 k
=====
Transaction Summary
=====
Install 4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm           1.4 MB/s | 209 kB    00:00
(2/4): telnet-0.17-85.el9.x86_64.rpm                  1.8 MB/s | 63 kB    00:00
(3/4): clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64.rpm 3.0 MB/s | 585 kB    00:00
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm               6.6 MB/s | 4.7 MB    00:00
=====
```

2.1 Установка Dovecot

- Переход в режим суперпользователя (`sudo -i`)
- Установка пакетов `dovecot` и `telnet` через `dnf`

```
[root@server.bahis.net ~]# dnf -y install dovecot telnet
Last metadata expiration check: 4:06:08 ago on Wed 11 Feb 2026 07:26:23 PM UTC.
Dependencies resolved.
=====
Package           Arch      Version                               Repository      Size
=====
Installing:
dovecot            x86_64    1:2.3.16-15.el9                     appstream      4.7 M
telnet             x86_64    1:0.17-85.el9                       appstream       63 k
Installing dependencies:
clucene-core       x86_64    2.3.3.4-42.20130812.e8e3d20git.el9  appstream     585 k
libexttextcat      x86_64    3.4.5-11.el9                        appstream     209 k
=====
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(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm    1.4 MB/s | 209 kB    00:00
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(3/4): clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86  3.0 MB/s | 585 kB    00:00
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm        6.6 MB/s | 4.7 MB    00:00
=====
```

2.1 Установка Dovecot

- Переход в режим суперпользователя (`sudo -i`)
- Установка пакетов `dovecot` и `telnet` через `dnf`
- Разрешение зависимостей и установка RPM-пакетов

```
[root@server.bahis.net ~]# dnf -y install dovecot telnet
Last metadata expiration check: 4:06:08 ago on Wed 11 Feb 2026 07:26:23 PM UTC.
Dependencies resolved.
=====
Package           Arch      Version                               Repository      Size
=====
Installing:
dovecot            x86_64    1:2.3.16-15.el9                     appstream      4.7 M
telnet             x86_64    1:0.17-85.el9                       appstream       63 k
Installing dependencies:
clucene-core       x86_64    2.3.3.4-42.20130812.e8e3d20git.el9  appstream      585 k
libexttextcat      x86_64    3.4.5-11.el9                        appstream      209 k
=====
Transaction Summary
=====
Install 4 Packages

Total download size: 5.6 M
Installed size: 20 M
Downloading Packages:
(1/4): libexttextcat-3.4.5-11.el9.x86_64.rpm           1.4 MB/s | 209 kB    00:00
(2/4): telnet-0.17-85.el9.x86_64.rpm                  1.8 MB/s | 63 kB    00:00
(3/4): clucene-core-2.3.3.4-42.20130812.e8e3d20git.el9.x86_64.rpm 3.0 MB/s | 585 kB    00:00
(4/4): dovecot-2.3.16-15.el9.x86_64.rpm               6.6 MB/s | 4.7 MB    00:00
=====
```

2.2 Настройка Dovecot

- В файле `/etc/dovecot/dovecot.conf` заданы протоколы `imap` и `pop3`

```
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
```

Рисунок 2: Настройка параметра `protocols` в `dovecot.conf`

2.2 Настройка Dovecot

- В файле `/etc/dovecot/dovecot.conf` заданы протоколы `imap` и `pop3`
- Разрешена работа сервиса по IMAP и POP3

```
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3

# A comma separated list of IPs or hosts where to listen in for connections.
```

Рисунок 2: Настройка параметра protocols в dovecot.conf

- В файле 10-auth.conf проверен параметр `auth_mechanisms = plain`

```
# Space separated list of wanted authentication mechanisms:  
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp  
#   gss-spnego  
# NOTE: See also disable_plaintext_auth setting.  
auth_mechanisms = plain  
##
```

Рисунок 3: Параметр `auth_mechanisms` в 10-auth.conf

- В файле 10-auth.conf проверен параметр `auth_mechanisms = plain`
- Подтверждён механизм аутентификации PLAIN

```
# Space separated list of wanted authentication mechanisms:  
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp  
#   gss-spnego  
# NOTE: See also disable_plaintext_auth setting.  
auth_mechanisms = plain  
##
```

Рисунок 3: Параметр `auth_mechanisms` в 10-auth.conf

- В файле `auth-system.conf.ext` указан `driver = pam`

```
# authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
passdb {
    driver = pam
    # [session=yes] [setcred=yes] [failure_show_msg=yes] [max_requests=<n>]
    # [cache_key=<key>] [<service name>]
    #args = dovecot
}
```

Рисунок 4: Параметры `passdb` и `userdb` в конфигурации Dovecot

- В файле `auth-system.conf.ext` указан `driver = pam`
- Для `userdb` используется `driver = passwd`

```
# authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
passdb {
    driver = pam
    # [session=yes] [setcred=yes] [failure_show_msg=yes] [max_requests=<n>]
    # [cache_key=<key>] [<service name>]
    #args = dovecot
}
```

Рисунок 4: Параметры `passdb` и `userdb` в конфигурации Dovecot

- В файле `auth-system.conf.ext` указан `driver = pam`
- Для `userdb` используется `driver = passwd`
- Используются системные учётные записи

```
# authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
passdb {
    driver = pam
    # [session=yes] [setcred=yes] [failure_show_msg=yes] [max_requests=<n>]
    # [cache_key=<key>] [<service name>]
    #args = dovecot
}
```

Рисунок 4: Параметры `passdb` и `userdb` в конфигурации Dovecot

```
userdb {  
  # <doc/wiki/AuthDatabase.Passwd.txt>  
  driver = passwd  
  # [blocking=no]  
  #args =  
  
  # Override fields from passwd  
  #override_fields = home=/home/virtual/%u  
}
```

Рисунок 5: Параметры passwd и userdb в конфигурации Dovecot

- В файле 10-mail.conf задано mail_location = maildir:~/Maildir

```
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
mail_location = maildir:~/Maildir
```

Рисунок 6: Настройка mail_location в конфигурации Dovecot

- В файле 10-mail.conf задано mail_location = maildir:~/Maildir
- Используется формат хранения Maildir

```
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
mail_location = maildir:~/Maildir
```

Рисунок 6: Настройка mail_location в конфигурации Dovecot

- В Postfix задан параметр `home_mailbox = Maildir/`

```
[root@server.bahis.net ~]# postconf -e 'home_mailbox = Maildir/'
[root@server.bahis.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd
audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb
bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-expor
ter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-
multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry doc
ker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-prox
y freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia
-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap i
maps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerbero
s kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-contro
l-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-service
s kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker
ldap ldaps libvirt libvirt-tls lighting-network llmnr llmnr-client llmnr-tcp llmnr-udp man
agesieve matrix mdns memcache minidlna mongod mosh moudn mqtt mqtt-tls ms-wbt mssql murmu
r mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemet
ry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pm
webapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2lin
k ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rootd rpc-b
ind rquotad rsh rsyncd rtsp salt-master samba,samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squ
id ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay sy
nergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp
-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery
-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-loc
```

Рисунок 7: Настройка параметра `home_mailbox` в Postfix

- В Postfix задан параметр `home_mailbox = Maildir/`
- Обеспечена совместимость доставки с Maildir

```
[root@server.bahis.net ~]# postconf -e 'home_mailbox = Maildir/'
[root@server.bahis.net ~]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd
audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb
bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-expor
ter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-
multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry doc
ker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-prox
y freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia
-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap i
maps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerbero
s kibana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-contro
l-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodeport-service
s kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker
ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp man
agesieve matrix mdns memcache minidlna mongod mosh moudm mqtt mqtt-tls ms-wbt mssql murmu
r mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemet
ry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pm
webapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2lin
k ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rootd rpc-b
ind rquotad rsh rsyncd rtsp salt-master samba,samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squ
id sssd ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay sy
nergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp
-client vdsml vnc-server warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery
-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-loc
```

Рисунок 7: Настройка параметра `home_mailbox` в Postfix

- В firewalld добавлены службы POP3, POP3S, IMAP, IMAPS

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 8: Настройка служб POP3 и IMAP в firewalld

- В firewalld добавлены службы POP3, POP3S, IMAP, IMAPS
- Выполнена перезагрузка конфигурации

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 8: Настройка служб POP3 и IMAP в firewalld

- В firewalld добавлены службы POP3, POP3S, IMAP, IMAPS
- Выполнена перезагрузка конфигурации
- Подтверждён список активных служб

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 8: Настройка служб POP3 и IMAP в firewalld

- Выполнена команда `restorecon -vR /etc`

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 9: Восстановление контекста SELinux

- Выполнена команда `restorecon -vR /etc`
- Восстановлены контексты SELinux

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 9: Восстановление контекста SELinux

- Выполнена команда `restorecon -vR /etc`
- Восстановлены контексты SELinux
- Обеспечена корректная работа служб

```
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.bahis.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.bahis.net ~]# firewall-cmd --reload
success
[root@server.bahis.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server.bahis.net ~]# restorecon -vR /etc
Relabeled /etc/resolv.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:net_conf_t:s0
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
```

Рисунок 9: Восстановление контекста SELinux

- Перезапущен Postfix

```
[root@server.bahis.net ~]# systemctl restart postfix
[root@server.bahis.net ~]# systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
[root@server.bahis.net ~]# systemctl start dovecot
```

Рисунок 10: Запуск и активация служб Postfix и Dovecot

- Перезапущен Postfix
- Активирована служба Dovecot через `systemctl`

```
[root@server.bahis.net ~]# systemctl restart postfix
[root@server.bahis.net ~]# systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
[root@server.bahis.net ~]# systemctl start dovecot
```

Рисунок 10: Запуск и активация служб Postfix и Dovecot

- Перезапущен Postfix
- Активирована служба Dovecot через `systemctl`
- Подтверждён успешный запуск служб

```
[root@server.bahis.net ~]# systemctl restart postfix
[root@server.bahis.net ~]# systemctl enable dovecot
Created symlink /etc/systemd/system/multi-user.target.wants/dovecot.service → /usr/lib/systemd/system/dovecot.service.
[root@server.bahis.net ~]# systemctl start dovecot
```

Рисунок 10: Запуск и активация служб Postfix и Dovecot

2.3 Проверка работы Dovecot

- Запущен мониторинг `tail -f /var/log/maillog`

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:51:13 server postfix/smtpd[10309]: disconnect from client.bahis.net[192.168.1.30]
ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:51:13 server postfix/local[10313]: 9A3D614A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:51:13 server postfix/qmgr[10270]: 9A3D614A9: removed
Feb 11 23:29:58 server postfix/postfix-script[1603]: starting the Postfix mail system
Feb 11 23:29:58 server postfix/master[1613]: daemon started -- version 3.5.25, configuratio
n /etc/postfix
Feb 11 23:44:53 server postfix/postfix-script[12040]: stopping the Postfix mail system
Feb 11 23:44:53 server postfix/master[1613]: terminating on signal 15
Feb 11 23:44:53 server postfix/postfix-script[12116]: starting the Postfix mail system
Feb 11 23:44:53 server postfix/master[12118]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 23:45:24 server dovecot[12166]: master: Dovecot v2.3.16 (7e2e900c1a) starting up for
imap, pop3
^C
[root@server.bahis.net ~]#
```

Рисунок 11: Мониторинг работы почтовых служб в журнале maillog

2.3 Проверка работы Dovecot

- Запущен мониторинг `tail -f /var/log/maillog`
- Зафиксирован запуск Postfix и Dovecot

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:51:13 server postfix/smtpd[10309]: disconnect from client.bahis.net[192.168.1.30]
ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:51:13 server postfix/local[10313]: 9A3D614A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:51:13 server postfix/qmgr[10270]: 9A3D614A9: removed
Feb 11 23:29:58 server postfix/postfix-script[1603]: starting the Postfix mail system
Feb 11 23:29:58 server postfix/master[1613]: daemon started -- version 3.5.25, configuratio
n /etc/postfix
Feb 11 23:44:53 server postfix/postfix-script[12040]: stopping the Postfix mail system
Feb 11 23:44:53 server postfix/master[1613]: terminating on signal 15
Feb 11 23:44:53 server postfix/postfix-script[12116]: starting the Postfix mail system
Feb 11 23:44:53 server postfix/master[12118]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 23:45:24 server dovecot[12166]: master: Dovecot v2.3.16 (7e2e900c1a) starting up for
imap, pop3
^C
[root@server.bahis.net ~]#
```

Рисунок 11: Мониторинг работы почтовых служб в журнале maillog

2.3 Проверка работы Dovecot

- Запущен мониторинг `tail -f /var/log/maillog`
- Зафиксирован запуск Postfix и Dovecot
- Подтверждена корректная работа служб

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# tail -f /var/log/maillog
Feb 11 20:51:13 server postfix/smtpd[10309]: disconnect from client.bahis.net[192.168.1.30]
ehlo=2 starttls=1 mail=1 rcpt=1 data=1 quit=1 commands=7
Feb 11 20:51:13 server postfix/local[10313]: 9A3D614A9: to=<bahis@server.bahis.net>, relay=
local, delay=0.01, delays=0.01/0/0/0, dsn=2.0.0, status=sent (delivered to mailbox)
Feb 11 20:51:13 server postfix/qmgr[10270]: 9A3D614A9: removed
Feb 11 23:29:58 server postfix/postfix-script[1603]: starting the Postfix mail system
Feb 11 23:29:58 server postfix/master[1613]: daemon started -- version 3.5.25, configuratio
n /etc/postfix
Feb 11 23:44:53 server postfix/postfix-script[12040]: stopping the Postfix mail system
Feb 11 23:44:53 server postfix/master[1613]: terminating on signal 15
Feb 11 23:44:53 server postfix/postfix-script[12116]: starting the Postfix mail system
Feb 11 23:44:53 server postfix/master[12118]: daemon started -- version 3.5.25, configurati
on /etc/postfix
Feb 11 23:45:24 server dovecot[12166]: master: Dovecot v2.3.16 (7e2e900c1a) starting up for
imap, pop3
^C
[root@server.bahis.net ~]#
```

Рисунок 11: Мониторинг работы почтовых служб в журнале maillog

- Выполнена команда MAIL=~/.Maildir mail

```
root@server.bahis.net ~]# MAIL=~/.Maildir mail  
s-nail: No mail for root at /root/.Maildir  
s-nail: /root/.Maildir: No such entry, file or directory
```

Рисунок 12: Проверка наличия почтового ящика пользователя root

- Выполнена команда MAIL=~ /Maildir mail
- Обнаружено отсутствие каталога /root/Maildir

```
root@server.bahis.net ~]# MAIL=~ /Maildir mail
s-nail: No mail for root at /root/Maildir
s-nail: /root/Maildir: No such entry, file or directory
```

Рисунок 12: Проверка наличия почтового ящика пользователя root

- Выполнена команда MAIL=~ /Maildir mail
- Обнаружено отсутствие каталога /root/Maildir
- Почтовый ящик root не создан

```
root@server.bahis.net ~]# MAIL=~ /Maildir mail  
s-nail: No mail for root at /root/Maildir  
s-nail: /root/Maildir: No such entry, file or directory
```

Рисунок 12: Проверка наличия почтового ящика пользователя root

- Выполнена команда `doveadm mailbox list -u bahis`

```
[root@server.bahis.net ~]# dovecmd mailbox list -u bahis
INBOX
```

Рисунок 13: Просмотр списка mailbox пользователя bahis

- Выполнена команда `doveadm mailbox list -u bahis`
- Отображён почтовый ящик INBOX

```
[root@server.bahis.net ~]# dovecadm mailbox list -u bahis
INBOX
```

Рисунок 13: Просмотр списка mailbox пользователя bahis

- Выполнена команда `doveadm mailbox list -u bahis`
- Отображён почтовый ящик INBOX
- Подтверждено создание mailbox пользователя

```
[root@server.bahis.net ~]# dovecmd mailbox list -u bahis
INBOX
```

Рисунок 13: Просмотр списка mailbox пользователя bahis

- На клиенте установлен пакет evolution

```
[root@client.bahis.net ~]# dnf -y install evolution
Last metadata expiration check: 4:01:41 ago on Wed 11 Feb 2026 07:57:14 PM UTC.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
evolution                     x86_64            3.40.4-11.el9_6.1  appstream         3.7 M
Installing dependencies:
evolution-langpacks          noarch            3.40.4-11.el9_6.1  appstream         5.6 M
highlight                     x86_64            3.60-5.el9         appstream          880 k
=====
Transaction Summary
=====
Install 3 Packages

Total download size: 10 M
Installed size: 56 M
Downloading Packages:
(1/3): highlight-3.60-5.el9.x86_64.rpm                3.4 MB/s | 880 kB    00:00
(2/3): evolution-langpacks-3.40.4-11.el9_6.1.noarch.rpm 5.6 MB/s | 5.6 MB    00:01
(3/3): evolution-3.40.4-11.el9_6.1.x86_64.rpm          3.6 MB/s | 3.7 MB    00:01
-----
Total                                                    8.0 MB/s | 10 MB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :                                                                 1/1
Installing : highlight-3.60-5.el9.x86_64                                1/3
Installing : evolution-langpacks-3.40.4-11.el9_6.1.noarch                2/3
Installing : evolution-3.40.4-11.el9_6.1.x86_64                         3/3
Running scriptlet: evolution-3.40.4-11.el9_6.1.x86_64                  3/3
Verifying   : evolution-3.40.4-11.el9_6.1.x86_64                        1/3
Verifying   : evolution-langpacks-3.40.4-11.el9_6.1.noarch              2/3
```

- На клиенте установлен пакет evolution
- Выполнена установка через dnf

```
[root@client.bahis.net ~]# dnf -y install evolution
Last metadata expiration check: 4:01:41 ago on Wed 11 Feb 2026 07:57:14 PM UTC.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
evolution                      x86_64            3.40.4-11.el9_6.1 appstream         3.7 M
Installing dependencies:
evolution-langpacks            noarch            3.40.4-11.el9_6.1 appstream         5.6 M
highlight                       x86_64            3.60-5.el9        appstream          880 k
=====
Transaction Summary
=====
Install 3 Packages

Total download size: 10 M
Installed size: 56 M
Downloading Packages:
(1/3): highlight-3.60-5.el9.x86_64.rpm                3.4 MB/s | 880 kB    00:00
(2/3): evolution-langpacks-3.40.4-11.el9_6.1.noarch.rpm 5.6 MB/s | 5.6 MB    00:01
(3/3): evolution-3.40.4-11.el9_6.1.x86_64.rpm          3.6 MB/s | 3.7 MB    00:01
-----
Total                                                    8.0 MB/s | 10 MB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :                                                                 1/1
Installing : highlight-3.60-5.el9.x86_64                                1/3
Installing : evolution-langpacks-3.40.4-11.el9_6.1.noarch                2/3
Installing : evolution-3.40.4-11.el9_6.1.x86_64                         3/3
Running scriptlet: evolution-3.40.4-11.el9_6.1.x86_64                   3/3
Verifying   : evolution-3.40.4-11.el9_6.1.x86_64                        1/3
Verifying   : evolution-langpacks-3.40.4-11.el9_6.1.noarch              2/3
```

- На клиенте установлен пакет evolution
- Выполнена установка через dnf
- Зависимости успешно разрешены

```
[root@client.bahis.net ~]# dnf -y install evolution
Last metadata expiration check: 4:01:41 ago on Wed 11 Feb 2026 07:57:14 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
evolution              x86_64           3.40.4-11.el9_6.1 appstream         3.7 M
Installing dependencies:
evolution-langpacks    noarch           3.40.4-11.el9_6.1 appstream         5.6 M
highlight              x86_64           3.60-5.el9        appstream         880 k
=====
Transaction Summary
=====
Install 3 Packages

Total download size: 10 M
Installed size: 56 M
Downloading Packages:
(1/3): highlight-3.60-5.el9.x86_64.rpm                3.4 MB/s | 880 kB    00:00
(2/3): evolution-langpacks-3.40.4-11.el9_6.1.noarch.rpm 5.6 MB/s | 5.6 MB    00:01
(3/3): evolution-3.40.4-11.el9_6.1.x86_64.rpm          3.6 MB/s | 3.7 MB    00:01
-----
Total                                                    8.0 MB/s | 10 MB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :                                                                 1/1
Installing : highlight-3.60-5.el9.x86_64                                1/3
Installing : evolution-langpacks-3.40.4-11.el9_6.1.noarch                2/3
Installing : evolution-3.40.4-11.el9_6.1.x86_64                         3/3
Running scriptlet: evolution-3.40.4-11.el9_6.1.x86_64                   3/3
Verifying   : evolution-3.40.4-11.el9_6.1.x86_64                        1/3
Verifying   : evolution-langpacks-3.40.4-11.el9_6.1.noarch              2/3
```

- В Evolution указаны имя `bahis` и адрес `bahis@bahis.net`

The screenshot shows the 'Identity' window in the Evolution email client. The window has a sidebar on the left with the following items: 'Welcome', 'Restore from Backup', 'Identity' (which is selected and highlighted), 'Receiving Email', 'Sending Email', 'Account Summary', and 'Done'. The main area of the window contains the following text and fields:

Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.

Required Information

Full Name:

Email Address:

Optional Information

Reply-To:

Organization:

Aliases:

At the bottom of the window, there is a checkbox labeled 'Look up mail server details based on the entered e-mail address' which is checked.

- В Evolution указаны имя `bahis` и адрес `bahis@bahis.net`
- Выбран режим ручной настройки

The screenshot shows the 'Identity' window in the Evolution email client. The window has a sidebar on the left with the following options: 'Welcome', 'Restore from Backup', 'Identity' (which is selected and highlighted), 'Receiving Email', 'Sending Email', 'Account Summary', and 'Done'. The main area of the window contains the following text and fields:

Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.

Required Information

Full Name:

Email Address:

Optional Information

Reply-To:

Organization:

Aliases:

At the bottom of the window, there is a checkbox labeled 'Look up mail server details based on the entered e-mail address' which is checked.

- Выбран протокол IMAP

Receiving Email

Welcome
Restore from Backup
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Server Type: IMAP

Description: For reading and storing mail on IMAP servers.

Configuration

Server: mail.bahis.net Port: 143

Username: bahis

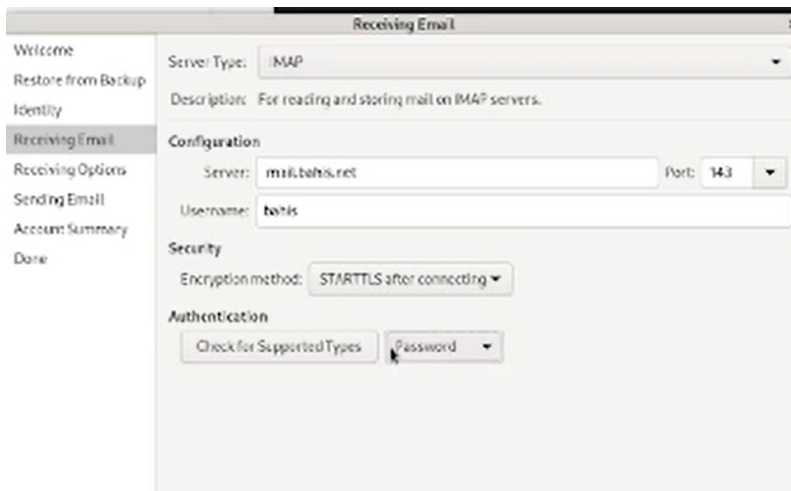
Security

Encryption method: STARTTLS after connecting

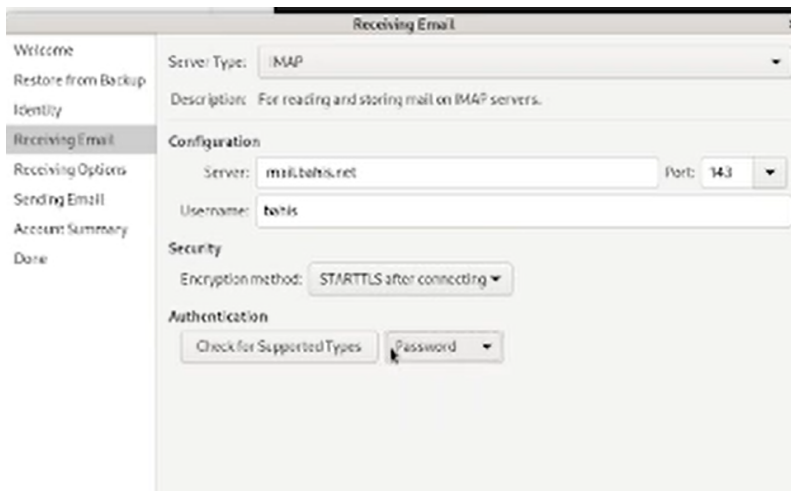
Authentication

Check for Supported Types Password

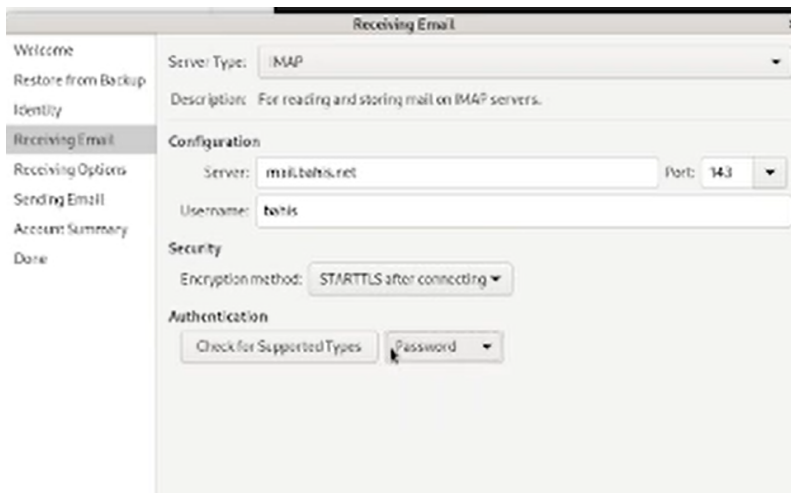
- Выбран протокол IMAP
- Сервер mail.bahis.net, порт 143



- Выбран протокол IMAP
- Сервер mail.bahis.net, порт 143
- Шифрование STARTTLS



- Выбран протокол IMAP
- Сервер mail.bahis.net, порт 143
- Шифрование STARTTLS
- Аутентификация — обычный пароль



- Указан SMTP-сервер mail.bahis.net, порт 25

Sending Email

Welcome
Restore from Backup
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Server Type: SMTP

Description: For delivering mail by connecting to a remote mailhub using SMTP.

Configuration

Server: mail.bahis.net Port: 25

☐ Server requires authentication

Security

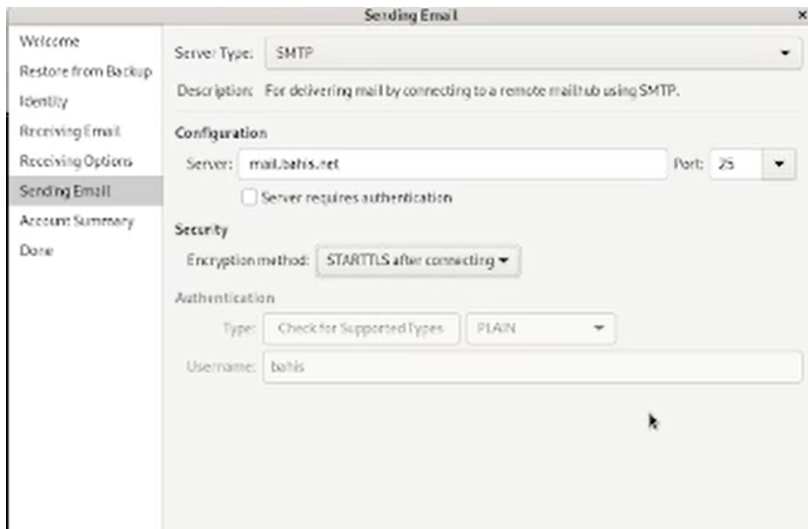
Encryption method: STARTTLS after connecting

Authentication

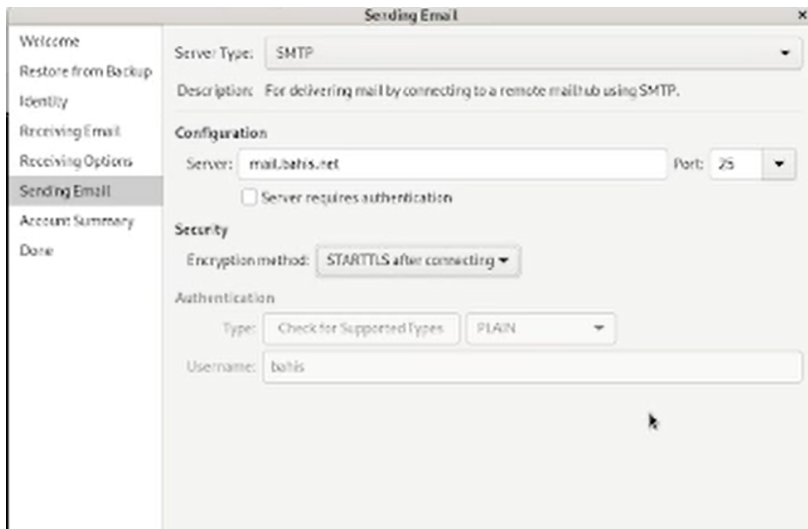
Type: Check for Supported Types PLAIN

Username: bahis

- Указан SMTP-сервер mail.bahis.net, порт 25
- Соединение без дополнительного шифрования



- Указан SMTP-сервер mail.bahis.net, порт 25
- Соединение без дополнительного шифрования
- Тип аутентификации — PLAIN



- Подтверждены параметры IMAP и SMTP

Account Summary

Welcome

Restore from Backup

Identity

Receiving Email

Receiving Options

Sending Email

Account Summary

Done

This is a summary of the settings which will be used to access your mail.

Account Information

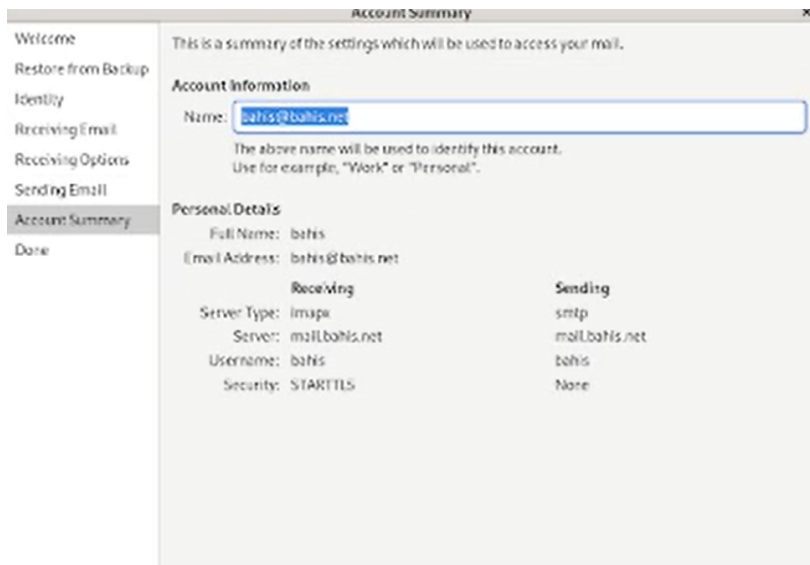
Name:

The above name will be used to identify this account.
Use for example, "Work" or "Personal".

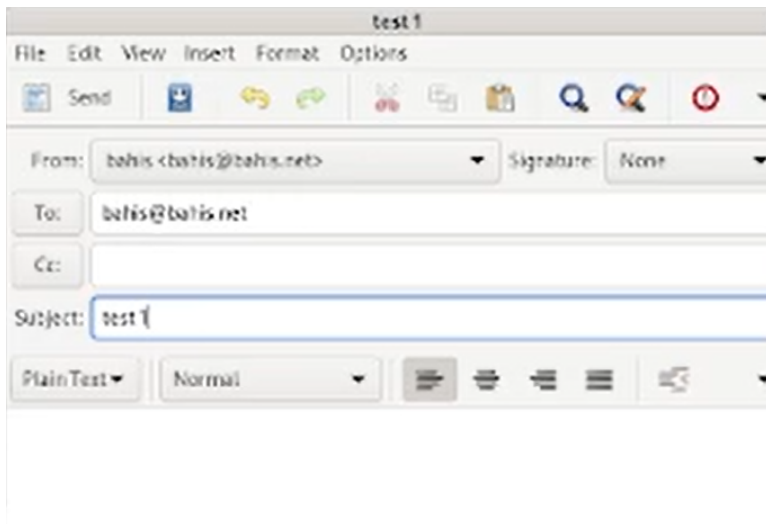
Personal Details

Full Name:	bahis	
Email Address:	bahis@bahis.net	
	Receiving	Sending
Server Type:	imap	smtp
Server:	mail.bahis.net	mail.bahis.net
Username:	bahis	bahis
Security:	STARTTLS	None

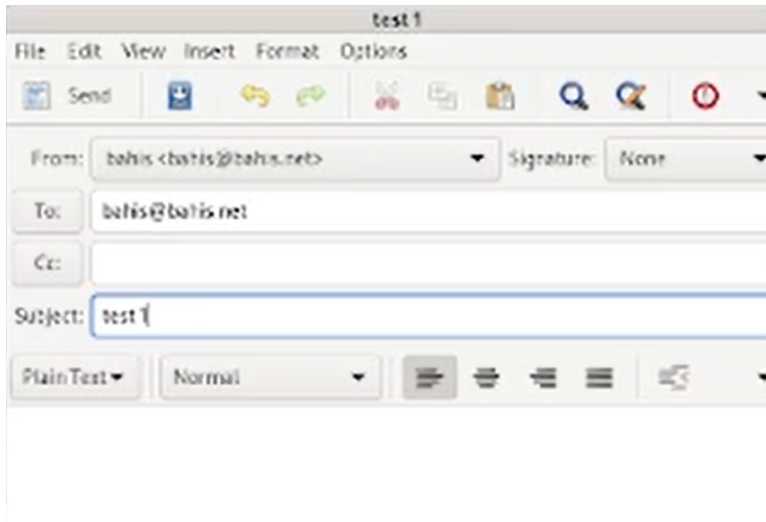
- Подтверждены параметры IMAP и SMTP
- Проверен сервер и пользователь



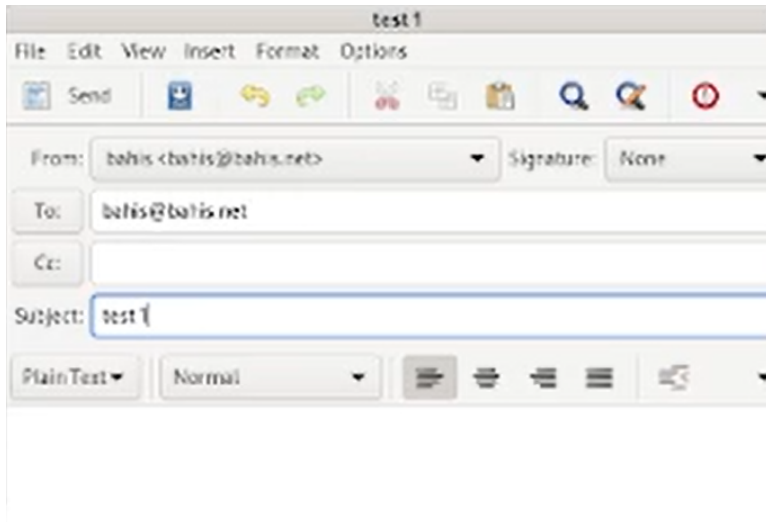
- Отправлены тестовые сообщения



- Отправлены тестовые сообщения
- Письма Test1, test2, test3 получены



- Отправлены тестовые сообщения
- Письма Test1, test2, test3 получены
- Подтверждена корректная доставка



- Выполнено подключение telnet mail.bahis.net 110

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7f7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```

- Выполнено подключение telnet mail.bahis.net 110
- Установлено соединение с Dovecot

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7f7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```

- Выполнено подключение telnet mail.bahis.net 110
- Установлено соединение с Dovecot
- Команда list отобразила три сообщения

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7f7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```

- Команда `retr 1` — получено письмо

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7ff7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```


- Команда `retr 1` — получено письмо
- Команда `dele 2` — письмо помечено к удалению

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7f7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```

- Команда `retr 1` — получено письмо
- Команда `dele 2` — письмо помечено к удалению
- Команда `quit` — завершение сеанса с удалением

```
+OK Logged in.  
list  
+OK 3 messages:  
1 620  
2 619  
3 619  
.  
retr  
-ERR Invalid message number:  
retr 1  
+OK 620 octets  
Return-Path: <bahis@bahis.net>  
X-Original-To: bahis@bahis.net  
Delivered-To: bahis@bahis.net  
Received: from client.bahis.net (client.bahis.net [192.168.1.30])  
        by server.bahis.net (Postfix) with ESMTTP id 446F425BE  
        for <bahis@bahis.net>; Thu, 12 Feb 2026 01:01:23 +0000 (UTC)  
Message-ID: <1be7f4e634d7806e8e7f7d25ce90499e111f3e5d.camel@bahis.net>  
Subject: test12  
From: bahis <bahis@bahis.net>  
To: bahis@bahis.net  
Content-Type: text/plain  
MIME-Version: 1.0  
Date: Thu, 12 Feb 2026 00:53:09 +0000  
User-Agent: Evolution 3.40.4 (3.40.4-11.el9_6.1)  
Content-Transfer-Encoding: 7bit  
  
send message from Client
```

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[root@server.bahis.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
```

Рисунок 22: Копирование конфигурационных файлов Dovecot в каталог provision

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создана структура `mail/etc/dovecot/conf.d`

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[root@server.bahis.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
```

Рисунок 22: Копирование конфигурационных файлов Dovecot в каталог provision

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создана структура `mail/etc/dovecot/conf.d`
- Скопированы файлы конфигурации Dovecot

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[root@server.bahis.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
```

Рисунок 22: Копирование конфигурационных файлов Dovecot в каталог provision

2.4 Внесение изменений в настройки внутреннего

- Переход в `/vagrant/provision/server`
- Создана структура `mail/etc/dovecot/conf.d`
- Скопированы файлы конфигурации Dovecot
- Подготовка к автоматическому провижинингу

```
[root@server.bahis.net ~]# cd /vagrant/provision/server
[root@server.bahis.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot/conf.d
[root@server.bahis.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/server/mail/etc/dovecot/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vagrant/provision/server/mail/etc/dovecot/conf.d/
[root@server.bahis.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/provision/server/mail/etc/dovecot/conf.d/
```

Рисунок 22: Копирование конфигурационных файлов Dovecot в каталог provision

- В mail.sh добавлена установка dovecot и telnet

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
```

- В mail.sh добавлена установка dovecot и telnet
- Открыты службы SMTP, POP3, POP3S, IMAP, IMAPS

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
```


- В mail.sh добавлена установка dovecot и telnet
- Открыты службы SMTP, POP3, POP3S, IMAP, IMAPS
- Задан home_mailbox = Maildir/

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
```

- В mail.sh добавлена установка dovecot и telnet
- Открыты службы SMTP, POP3, POP3S, IMAP, IMAPS
- Задан home_mailbox = Maildir/
- Выполнены restorecon, перезапуск Postfix и запуск Dovecot

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
```

- В mail.sh добавлена установка dovecot и telnet
- Открыты службы SMTP, POP3, POP3S, IMAP, IMAPS
- Задан home_mailbox = Maildir/
- Выполнены restorecon, перезапуск Postfix и запуск Dovecot
- Реализована автоматическая настройка сервера

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=smtp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
```

- В `client/mail.sh` добавлена установка `evolution`



```
root@client:/vagrant/provision/client
GNU nano 5.6.1 mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
dnf -y install evolution
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
```

Рисунок 24: Изменение скрипта `mail.sh` на клиенте

- В `client/mail.sh` добавлена установка `evolution`
- Автоматизирована настройка клиентской части



```
root@client:/vagrant/provision/client
GNU nano 5.6.1 mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
dnf -y install evolution
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
```

Рисунок 24: Изменение скрипта `mail.sh` на клиенте

Раздел 3

3. Выводы

3.1 Выводы

- Установлены и настроены Postfix и Dovecot

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3
- Настроен firewalld и восстановлены контексты SELinux

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3
- Настроен firewalld и восстановлены контексты SELinux
- Выполнена проверка через журнал, mail, doveadm и Telnet

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3
- Настроен firewalld и восстановлены контексты SELinux
- Выполнена проверка через журнал, mail, doveadm и Telnet
- Подтверждена работа клиента Evolution

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3
- Настроен firewalld и восстановлены контексты SELinux
- Выполнена проверка через журнал, mail, doveadm и Telnet
- Подтверждена работа клиента Evolution
- Реализована интеграция конфигурации в Vagrant

3.1 Выводы

- Установлены и настроены Postfix и Dovecot
- Настроена доставка почты в формате Maildir
- Реализована аутентификация через PAM и passwd
- Обеспечена работа IMAP и POP3
- Настроен firewalld и восстановлены контексты SELinux
- Выполнена проверка через журнал, mail, doveadm и Telnet
- Подтверждена работа клиента Evolution
- Реализована интеграция конфигурации в Vagrant
- Обеспечена воспроизводимость почтовой системы